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Plasma and Gamma-Globulin Infusion in Chronic Inflammatory Polyneuropathy

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SUMMARY

We treated 17 patients with chronic inflammatory polyneuropathy (CIP) with infusions of fresh-frozen plasma (FFP). Thirteen patients had a beneficial response: 12 improved considerably and one moderately. The first signs of improvement were seen within 8 days of the onset of treatment. None of these patients had had spontaneous remissions for 2 months prior to treatment, and none had received immunosuppressive medication during the plasma infusions. The treatment effect was short-lasting in 9 patients. Re-institution of the FFP infusions was always followed by improvement. The same effect could be achieved by intravenous gamma-globulin.

We conclude that infusions with FFP, of which the IgG fraction is the effective part, may induce a rapid and clinically important improvement in patients with CIP.

Key words: *Chronic inflammatory polyneuropathy – Fresh-frozen plasma infusion – Intravenous gamma-globulin*

INTRODUCTION

The chronic inflammatory polyneuropathies (CIP) are characterized by progressive muscle weakness and usually slight sensory disturbances (Dyck et al. 1975). These neuropathies may be preceded by a viral infection. The cerebrospinal fluid protein is

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elevated at some stage during the course of the disease, and the nerve conduction velocities are slowed (Prineas and McLeod 1976). Sural nerve biopsies may show demyelination and mononuclear cell infiltrates (Dyck et al. 1975). Evidence of systemic disease is lacking. Corticosteroids are often effective (Dyck et al. 1982), and case reports suggest that azathioprine (Pentland 1980) and cyclophosphamide (Dalakas and Engel 1981) may also be beneficial. Plasmapheresis has been reported to produce dramatic improvement (Server et al. 1979).

Autoimmune thrombocytopenia is a disorder which may occur after viral infections; it can have an acute or a chronic course, and may respond to corticosteroids and to plasmapheresis. In addition, autoimmune thrombocytopenia may respond to infusion of fresh-frozen plasma (FFP) and to intravenous gamma-globulin (Imbach et al. 1981; Newland et al. 1983). Therefore, we investigated whether infusions of FFP and gamma-globulins were also effective in the chronic inflammatory polyneuropathy. The beneficial effect of FFP in two CIP patients has been reported (Busch et al. 1982). Since that time 15 CIP patients have been treated with FFP. In 9 of these patients intravenous gamma-globulin was administered.

MATERIAL AND METHODS

Patients

Patients with CIP were treated with FFP in two University Departments of Neurology (Rotterdam Dijkzigt and Amsterdam AMC). Chronic inflammatory polyneuropathy was defined as a sensorimotor polyneuropathy in the absence of diabetes, uraemia, paraproteinaemia, connective tissue disease, porphyria, heavy metal intoxication and malignancy. The family history had to be negative for hereditary polyneuropathy.

The progression of weakness exceeded a duration of 4 weeks, and/or had a relapsing course. All patients underwent electromyographic investigation according to standard methods and at least one lumbar puncture.

Treatment

FFP infusions were considered in all CIP patients admitted between March 1980 and June 1984, and administered if the disability of the patient was such that normal daily activities could not be carried out.

We infused 0.1 l of FFP/kg body weight within 5–7 days. These infusions were repeated when improvement ceased. If patients needed repeated FFP infusions to prevent secondary deterioration, the lowest amount and frequency of FFP were determined. In patients who required frequent infusions, we investigated whether the administration of intravenous gamma-globulin was also effective. This treatment was started after the FFP infusions had been discontinued and the patient had deteriorated. Intravenous gamma-globulin (human immunoglobulin for intravenous use, Central Laboratory Dutch Red Cross, Amsterdam) was initially given at a dose of 0.4 g/kg body weight/day for 5 consecutive days. Thereafter the lowest effective dosage with the largest possible intervals without deterioration of the patient, was empirically found.

Assessment of therapy effect

The disability of these patients was estimated by recording only the function of the legs.

Grade 1: Cannot move legs.

2: Moves legs but is unable to walk with assistance.

3: Walks with assistance.

4: Walks without assistance but has weakness and/or sensory disturbances.

5: No abnormalities.

Assessments were made daily during the treatment and at least 1 month after the onset of the infusions. Physicians, who examined the patients knew which treatment had been given.

RESULTS

We treated 17 patients with FFP infusions. Age, CSF total protein, reduction of peroneal nerve conduction velocity and the responses to treatment are summarized in

TABLE 1
TREATMENT RESPONSE IN 17 CIP PATIENTS

Patients			CSF total protein ^b (g/l)	Reduction motor NCV ^a peroneal nerve (%)	Response to treatment		
No.	Female/Male	Age (yr)			FFP	Intravenous gammaglobulin	Other
1	m	60	0.35	45	+		
2	f	59	1.44	55	+	+	
3	f	42	0.47	15	+	+	
4	m	14	0.97	20	+	+	C ⁻ A ⁻
5	f	32	1.19	64	+	+	
6	m	57	0.61	77	+	+	
7	m	11	0.41	40	+	+	
8	m	55	1.20	77	+	+	
9	m	52	1.07	43	+		C ⁺ A ⁺
10	m	45	1.22	38	+		C ⁻
11	m	60	1.62	60	+		
12	m	31	0.91	43	+	+	C ⁺ A ⁺
13	m	63	0.47	43	+		C ⁻
14	m	36	3.00	100	-		
15	m	71	0.65	32	-		C ⁻
16	m	54	1.90	19	-		C ⁺
17	m	16	0.53	100	-	-	

^a NCV: nerve conduction velocity.

^b Normal values up to 0.60.

+, improvement; -, no improvement; C, prednisone; A, azathioprine; C and A were not given in combination with FFP or gamma-globulins.

Table 1. In 5 patients the CSF total protein level was not raised. Lumbar puncture was not repeated in these patients. The CSF cell count did not exceed 7 in any of the patients.

Response to treatment

Nine of the 12 patients in Rotterdam and 4 of the 5 patients in Amsterdam showed a beneficial response to FFP. In retrospect we could not identify features that distinguished patients with and without response to FFP. Prednisone was administered at a dose of 60 mg a day for at least 6 weeks in 7 patients. Four did not respond to this treatment. Three of these four improved after FFP treatment. One patient neither improved after prednisone nor after FFP. One patient without response to FFP, improved after the administration of prednisone. Intravenous gamma-globulin was given to 9 patients. Eight responded to this treatment and had had a beneficial response to FFP. Prednisone or other immunosuppressive treatment was not given during FFP and gamma-globulin infusion and had not been given for 2 months prior to the first infusions.

Effect of FFP treatment

The effect of FFP treatment in 13 patients was rapid and functionally significant. These 13 patients had had no improvement for at least 2 months before the FFP treatment started. Ten of these had never had spontaneous remissions.

The first effects of treatment were always seen within 8 days of the onset of therapy. The treatment effects are summarized in Table 2. Patients 2, 4 and 7 needed more than one course of FFP within one month to reach grade 3. The treatment effect was considerable in patients 1–12. The improvement in patient 13 was obvious, although it could not be expressed with our grading system. He was unable to climb stairs without support before treatment, but after treatment he could.

After 1 month and with repeated courses of FFP, further improvement was seen in patients 1, 4 and 7; they reached grades 5, 4 and 4, respectively.

Four patients (9, 10, 11 and 13) remained stable for at least 4 months after one course of treatment. One of these patients is described in case history 1. The effect was short-lasting in 9 patients. In these the first sign of deterioration was seen, on average, 3 weeks after the last infusion. When deterioration started, this was rapidly progressive. One of these patients is described in case history 2. Reinstitution of FFP treatment resulted in improvement in all. None of the patients became refractory after repeated courses of treatment. The frequency and dosage varied from 4 l FFP once a week to 3 l once in 3 weeks. One patient reached remission (follow-up 14 months) after 1 year of intermittent infusions. The remaining 8 patients were dependent on intermittent infusions for a period of at least 2 years. In these 8, intravenous gamma-globulins were administered.

Effect of FFP treatment on nerve conduction velocity

In 6 patients showing a beneficial response to treatment, the motor nerve conduction velocity and the distal motor latency of the right median nerve were measured

TABLE 2

13 PATIENTS WITH CIP AND A BENEFICIAL RESPONSE TO FFP INFUSION

Pat. no.	Duration of weakness without spontaneous improvement previous to therapy (months)	Grade of disability		First signs of improvement after onset of treatment (days)
		at onset of treatment	one month after treatment	
1 ^a	3	3	4	5
2	7	2	3	5
3	9	3	4	7
4 ^a	2	1	3	4
5	2	3	4	7
6	12	2	4	7
7 ^a	3	1	3	3
8	9	2	4	3
9	12	2	4	3
10	4	3	4	3
11	3	3	4	3
12	2	3	5	3
13 ^b	5	4	4	7

Grade 1: Cannot move legs.

2: Moves legs but is unable to walk with assistance.

3: Walks with assistance.

4: Walks without assistance but has weakness and/or sensory disturbances.

5: No abnormalities.

^a These patients showed further improvement after one month; patient 1 reached grade 5, patient 4 grade 4 and patient 7 grade 4.

^b This patient was the least disabled patient of this group. In the functional scale of the legs there was no change, but there was a clear improvement as he was not able to climb stairs without support before treatment, which he could after treatment.

before treatment and after treatment, when improvement had reached its maximum. The nerve conduction velocity was measured over the forearm and recorded from the abductor pollicis brevis with surface electrodes. The results are shown in Table 3.

In all these 6 patients the grip strength as measured with a dynamometer had increased. In 2 patients (1, 3) the distal motor latency returned to within normal limits. A decrease of latency was observed in 4 patients (2, 4, 5, 6). The nerve conduction velocity over the forearm returned to normal values in 1 patient (1) and increased in 3 patients (2, 4, 6). One patient, however, had a decrease in nerve conduction velocity despite improvement of grip strength.

Effects of intravenous gamma-globulin

In 9 patients treated with intravenous gamma-globulin, the response to treatment was the same as after FFP infusions. In 8, the first signs of improvement were seen within 8 days of the onset of therapy. The required amount and frequency of treatment varied from 27 g once a week to 27 g once in 3 weeks. Discontinuation led to deterio-

TABLE 3

MOTOR NERVE CONDUCTION VELOCITY AND DISTAL MOTOR LATENCY BEFORE AND AFTER FFP INFUSION

These 6 patients showed an increase in grip strength. d.m.l. = distal motor latency; n.c.v. = nerve conduction velocity.

Patient No.	Before treatment		After treatment	
	d.m.l. (m.s)	n.c.v. (m/s)	d.m.l. (m.s)	n.c.v. (m/s)
1	8.4	46	3.3	54
2	6.1	11	5.4	20
3	4.0	38	3.6	39
4	13.5	16	8.2	36
5	7.6	30	7.2	19
6	7.2	0	5.4	23

ration and re-institution was always followed by improvement. None of these patients reached a long-lasting remission. The longest duration of this treatment was 2 years; the shortest 6 months. One patient did not respond to intravenous gamma-globulin. This patient had not improved after FFP.

Side-effects

Allergic reactions such as skin rash were frequently noted in patients treated with FFP. These reactions were never severe and disappeared rapidly after the intravenous administration of Clemastin. Two patients developed non-A-non-B hepatitis which had a benign course in both.

Allergic reactions after intravenous gamma-globulin were not observed. The only side-effect was a slight headache for a few hours after the infusion in some patients.

Case history 1

This 60-year-old man developed progressive weakness in legs and arms and tingling in legs and hands. Four months after the onset of these symptoms he was referred to the AMC Amsterdam. On admission he could not walk without assistance. Power was reduced in arms (MRC grade 4) and legs (proximal MRC grade 4, distal 0). Light touch was impaired in the forearms and legs and position sense was reduced in the feet. There was universal tendon areflexia. The CSF protein was elevated to 1.62 g/l. Motor nerve conduction velocities in the right median nerve on recording from abductor pollicis brevis was reduced over the forearm (20 m/s) with a prolonged distal motor latency (9.0 m.s). There was a reduced motor nerve conduction velocity in the peroneal nerve (21 m/s). After infusion of 7 l FFP, the power in the limbs improved. The first signs of improvement could be noted 3 days after initiation of the infusions. Eleven days after the start of treatment he had only slight weakness of the distal arm muscles, no weakness of the proximal leg muscles and the foot elevators had improved up to MRC

grade 4. One month after the first infusion there was only slight weakness of the foot elevators. Four months later he showed gradual deterioration. After progression for a period of 3 months, he was re-admitted. On examination there was a glove and stocking sensory loss for touch. The power was reduced in his arms (distal MRC grade 4) and in his legs (proximal MRC grade 4 and distal grade 2). After 7 l of FFP, the improvement started 6 days after the first infusion. One month later he had only slight weakness of the footelevators. There has been no further deterioration (follow-up 3.5 years).

Case history 2

This 14-year-old boy was admitted to the Dijkzigt Hospital after a 2-month history of progressive weakness in the arms and legs. On examination he could not lift his arms off the bed and was unable to squeeze. His legs were paralyzed (MRC grade 0). There was areflexia, except for one hypo-active ankle reflex. Sensation was unimpaired. Motor nerve conduction velocities in the right peroneal nerve (40 m/s) and the right median nerve (41 m/s) were reduced. The H-reflex could not be elicited on either side. The CSF protein was elevated at 0.97 g/l. A right sural nerve biopsy showed widening of the subperineurial space and signs of demyelination and remyelination. There were no signs and symptoms of an associated disease. He received 6 l of FFP on 6 consecutive days, followed by another 6 l 14 days later. One month after the onset of FFP infusions he was able to walk with assistance and the power in his arms had increased up to MRC grade 3. His condition remained stable for 8 days. Subsequently the power of his arms and legs decreased. Prednisone was started at 60 mg/day and 3 weeks later azathioprine (200 mg/day) was added. The deterioration continued and after 8 weeks of prednisone treatment his condition was just as bad as on admission. For 3 months no treatment was given and no change in his condition was noted. Subsequently, plasmapheresis was started whereby 4 l of plasma were replaced weekly by albumin. This resulted in a rapid improvement, comparable to the effect of FFP infusion. After 3 months the plasmapheresis was discontinued, followed by a rapid deterioration. Improvement could be achieved by repeated FFP infusions. He could walk without assistance and the power in his arms had increased to MRC grade 4. The FFP infusions were discontinued and prednisone, 60 mg/day, was started. This resulted in a rapid deterioration. Six weeks after the onset of prednisone administration he walked with assistance and the power in his arms had decreased to MRC grade 2. The prednisone was tapered off and replaced by FFP infusions. This was followed by a rapid improvement and after repeated infusions he had only slight weakness of the distal arm muscles and footelevators. This condition could be maintained by infusion of 2 l of FFP twice a week. Discontinuation of these FFP infusions was again followed by a decrease in power. After 3 weeks the power of his distal arm muscles and footelevators had decreased to MRC grade 3. Infusion of gamma-globulin, 27 g/day, on 5 consecutive days resulted in improvement. Two weeks later he had only slight weakness of the footelevators and distal arm muscles. This condition could be maintained by weekly infusion of 27 g gamma-globulin. He was able to attend school and the infusions were given weekly within a 2.5 h period.

After 4 months the infused amount was diminished to 12 g. This was followed

by a deterioration. Improvement could be achieved by re-institution of the former dosage. This was repeated several times. At the onset of one of these attempts to diminish the dosage of gamma-globulin, the nerve conduction velocity of the right median nerve was 46 m/s over the forearm with a distal motor latency of 4.7 m.s.

Three weeks after the gamma-globulin had been diminished, the nerve conduction velocity was 38 m/s and the distal motor latency 6.8 m.s. Three weeks after re-institution of 27 g gamma-globulin the nerve conduction velocity was 44 m/s and the distal motor latency 4.8 m.s. After 2 years of weekly infusions this patient has only slight weakness of distal arm muscles and foot elevators.

DISCUSSION

Thirteen of 17 patients with CIP improved rapidly after infusion of FFP. The effect was considerable in 12 and moderate in one. For three reasons, we attribute these improvements to FFP. Firstly, signs of improvement were noted very soon after the onset of treatment, while no spontaneous improvement had occurred for at least 2 months prior to this treatment. Ten patients had never even had spontaneous remissions during the course of their disease. Secondly, the rapid increase in muscle strength contrasted with the more gradual increase which may occur during the natural course. Thirdly, 9 patients deteriorated on average 3 weeks after the FFP infusions and improvement could repeatedly be achieved by re-institution of the infusions. This short-lasting effect of 3 weeks is similar to the effect of FFP in autoimmune thrombocytopenia (Newland et al. 1983) and in myasthenia gravis (Gajdos et al. 1984).

Nine patients required repeated infusions to prevent deterioration after initial improvement. This could be achieved by infusion of FFP varying from 4 l once a week to 3 l once in 3 weeks. For these infusions the patients had to be repeatedly hospitalized. In patients with autoimmune thrombocytopenia, both FFP and gamma-globulin infusions are effective (Newland et al. 1983). The advantage of gamma-globulin infusions is that the same amount of gamma-globulin is present in a smaller volume when compared with FFP, and therefore administration in an outpatient department is possible.

All the CIP patients who needed repeated infusions of FFP to maintain their improved neurological condition, responded to intravenous gamma-globulin. Several times this intermittent therapy was discontinued to investigate whether the patient had reached a permanent remission. When the patient deteriorated, improvement could invariably be achieved by re-institution of the therapy. None became refractory after several courses of infusions, which is in contrast with the experience in autoimmune thrombocytopenia, as in these patients the response to gamma-globulin may gradually decrease (Imbach et al. 1981).

FFP treatment may also be effective in the acute inflammatory polyneuropathies. We have treated 12 AIP patients with FFP infusions; two showed such dramatic improvement that it was felt that the improvement was caused by this treatment. However, treatment effects in these acute polyneuropathies are difficult to assess in an uncontrolled series.

Since patients with CIP may respond to both FFP and gamma-globulin infusions, it is likely that the IgG fraction of FFP is effective because the infused gamma-globulin consisted mainly of monomeric IgG. The immunosuppressive effect of IgG infusions in patients with CIP may be compared with the effect in patients with autoimmune thrombocytopenia. It has been suggested that IgG infusions are effective by blockade of the Fc-receptors preventing the elimination of antibody-coated platelets (Salama et al. 1984). Such a mechanism might play a role in patients with CIP as during the course of the disease mononuclear cell infiltrates are present in biopsies of the sural nerves. Blockade of these cells might prevent the cytopathological effect on the peripheral nerve.

Another hypothesis explaining the effectiveness of IgG in CIP is that anti-idiotypic antibodies, present in the infused IgG, bind to auto-antibodies against peripheral nerve tissue antigens. This mechanism is the likely explanation of the effect of IgG infusion in patients with auto-antibodies to factor VIII (Sultan et al. 1984).

A third possibility is that receptors on the peripheral nerve involved in the nerve conduction have homology with immunoglobulin structures. These receptors and the infused immunoglobulins may compete for binding with the auto-antibodies responsible for nerve conduction disturbances.

In 6 patients with a beneficial response to treatment the measurements of the nerve conduction velocity and the distal motor latency of the median nerve were repeated after treatment. The distal motor latency decreased in all and the nerve conduction velocity increased in 4 patients. In one patient there was a decrease of nerve conduction velocity despite significant clinical improvement. Slowed nerve conduction velocities, however, do not explain weakness. The weakness in CIP can better be explained by an axonal defect and as the improvement of weakness was remarkably fast after FFP or gamma-globulin infusions the axonal defect is probably functional rather than anatomical. The malfunction of axons could be caused by a mononuclear cell secretion factor or an antibody and if the secretion of that factor is inhibited or if the antibody is bound, this functional defect may be rapidly reversed, resulting in an increase of muscle power.

In patients with CIP, the infusion of gamma-globulins is an alternative to treatment with corticosteroids or plasmapheresis. The disadvantage of corticosteroids is that high doses are required for a long period with the risk of serious side-effects. In addition, intravenous gamma-globulin may be effective in patients who do not respond to steroid treatment. The disadvantage of plasmapheresis is that special technical equipment is required and the procedure is expensive. Not seldom, an A-V fistula is needed which may have secondary complications such as thrombosis, or bleeding due to anticoagulant treatment. Furthermore, the 'shunt' may cause limitations in daily activities. The disadvantages of FFP and gamma-globulin infusion are the risks of transmittable diseases such as hepatitis and AIDS. After FFP infusions, two of our patients developed a non-A-non-B hepatitis. The risk of hepatitis is probably less if, instead of FFP, intravenous gamma-globulins are used, although recently non-A-non-B hepatitis has been reported in agammaglobulinaemic patients after intravenous gamma-globulin treatment (Lever et al. 1984). To date, AIDS has not been reported following intravenous gamma-globulin treatment.

Intravenous gamma-globulin and FFP are advances in the management of CIP patients who are severely disabled. The high costs of intravenous gamma-globulin treatment concur easily with the costs of long-lasting hospitalization which is needed if severely disabled patients are treated with corticosteroids.

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